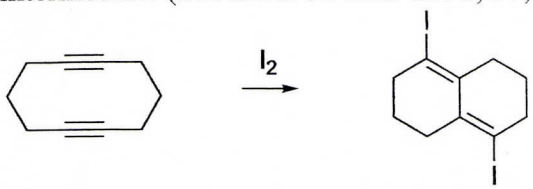
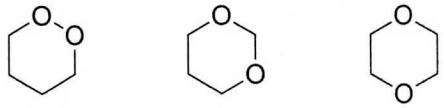
ORGANIC CUME 01-26-2018

1. Halogenation of alkynes with Cl_2 or Br_2 can generally be achieved with high yields, while halogenation of alkynes with I_2 typically gives low yields. However, the following reaction is successfully completed with I_2 in high yields (94%) to afford a potentially useful functionalized and conformationally constrained diiododiene synthetic intermediate (*Tetrahedron Lett.* **1996**, *37*, 1571-1574).



(a) The first step of the process likely involves one of the triple bonds attacking I_2 to give a bridged iodonium ion. Draw this intermediate.
(b) In the second step of the mechanism, it is unlikely that the other triple bond attacks the iodonium ion in an intramolecular nucleophilic attack, because the resulting carbocation would be too high in energy. Draw this carbocation, and explain why it is too high in energy to form.
(c) In the second step of the mechanism, it is more likely that an iodide ion attacks the triple bond, which simultaneously attacks the iodonium ion (in a concerted fashion). Therefore, the entire mechanism has just two steps. Draw both steps of this mechanism, as described.

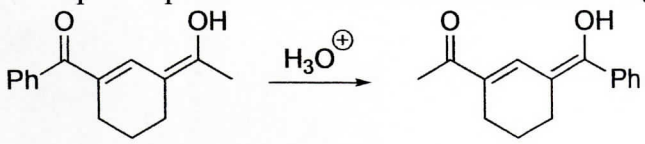
2. Consider the three constitutional isomers of dioxane ($C_4H_8O_2$):



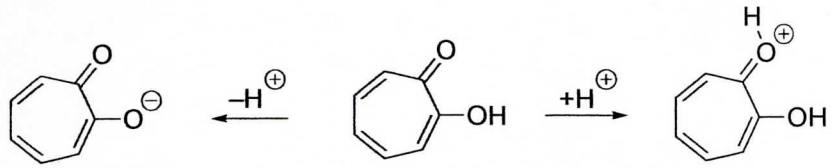
1,2-Dioxane 1,3-Dioxane 1,4-Dioxane

One of these constitutional isomers is stable under basic conditions as well as mildly acid conditions and is therefore used as a common solvent. Another isomer is only stable under basic conditions but undergoes hydrolysis under mildly acidic conditions. The remaining isomer is extremely unstable and potentially explosive. Identify each isomer and explain the properties of each compound.

3. Propose a plausible mechanism for the following transformation.



4. Tropolone **1** is a compound that is both fairly acidic and fairly basic. It is acidic because it is capable of losing a proton (H^+) to form a relatively stable anion **2**, while it is basic because of its ability to receive a proton to form cation **3** (*J. Org. Chem.* **1997**, *62*, 3200-3207).



2 **1** **3**

(a) Draw all significant resonance structures of anion **2** and of cation **3** and explain why each of these ions is stabilized.

(b) In compound **1**, the bond length for the C-C bonds vary greatly and alternate in length (long, short, long, etc.). But in ions **2** and **3**, the lengths of the bonds are similar. Explain.

(c) The intramolecular hydrogen bonding interaction in **3** is significantly diminished in comparison to **1**. Explain.